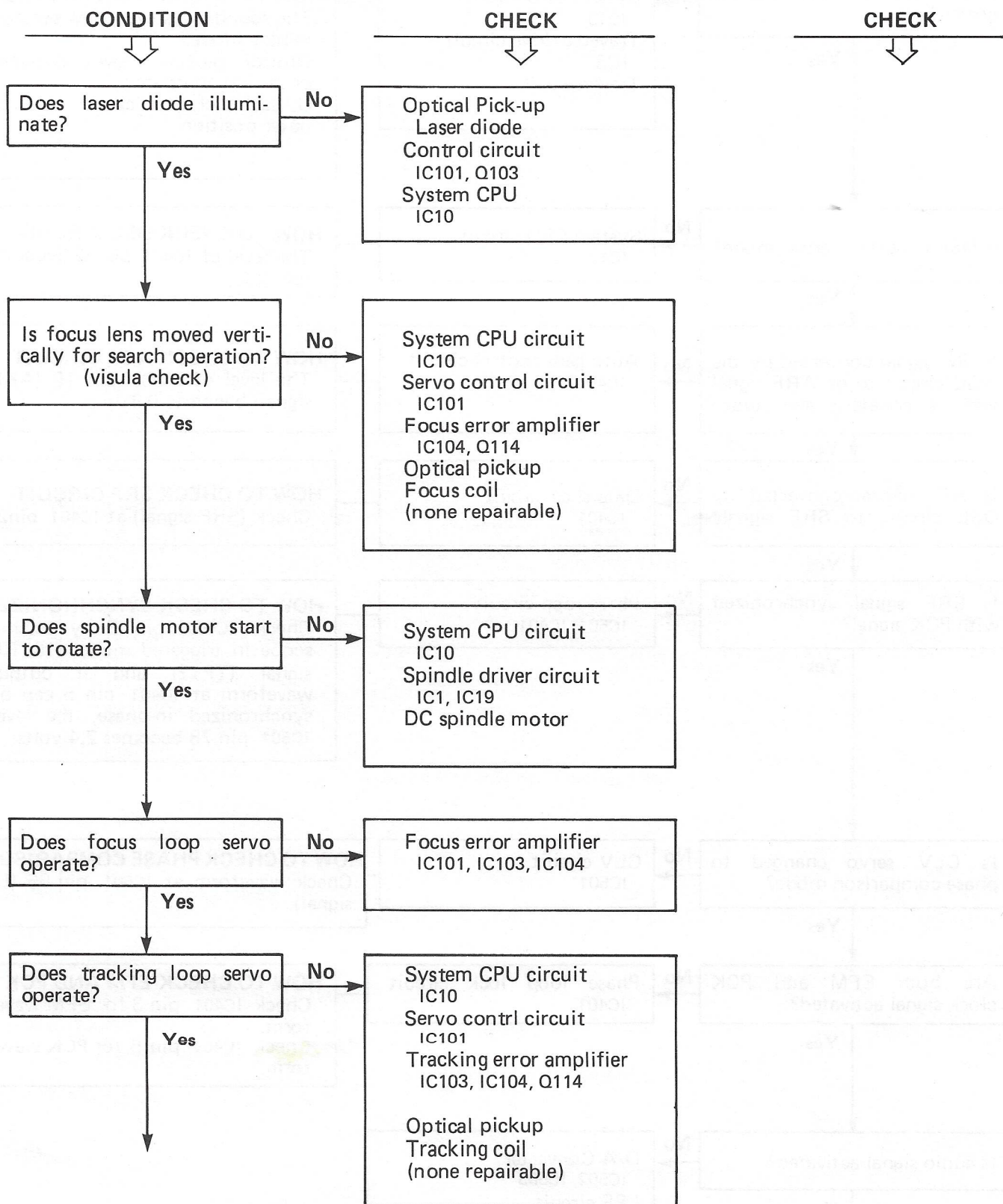
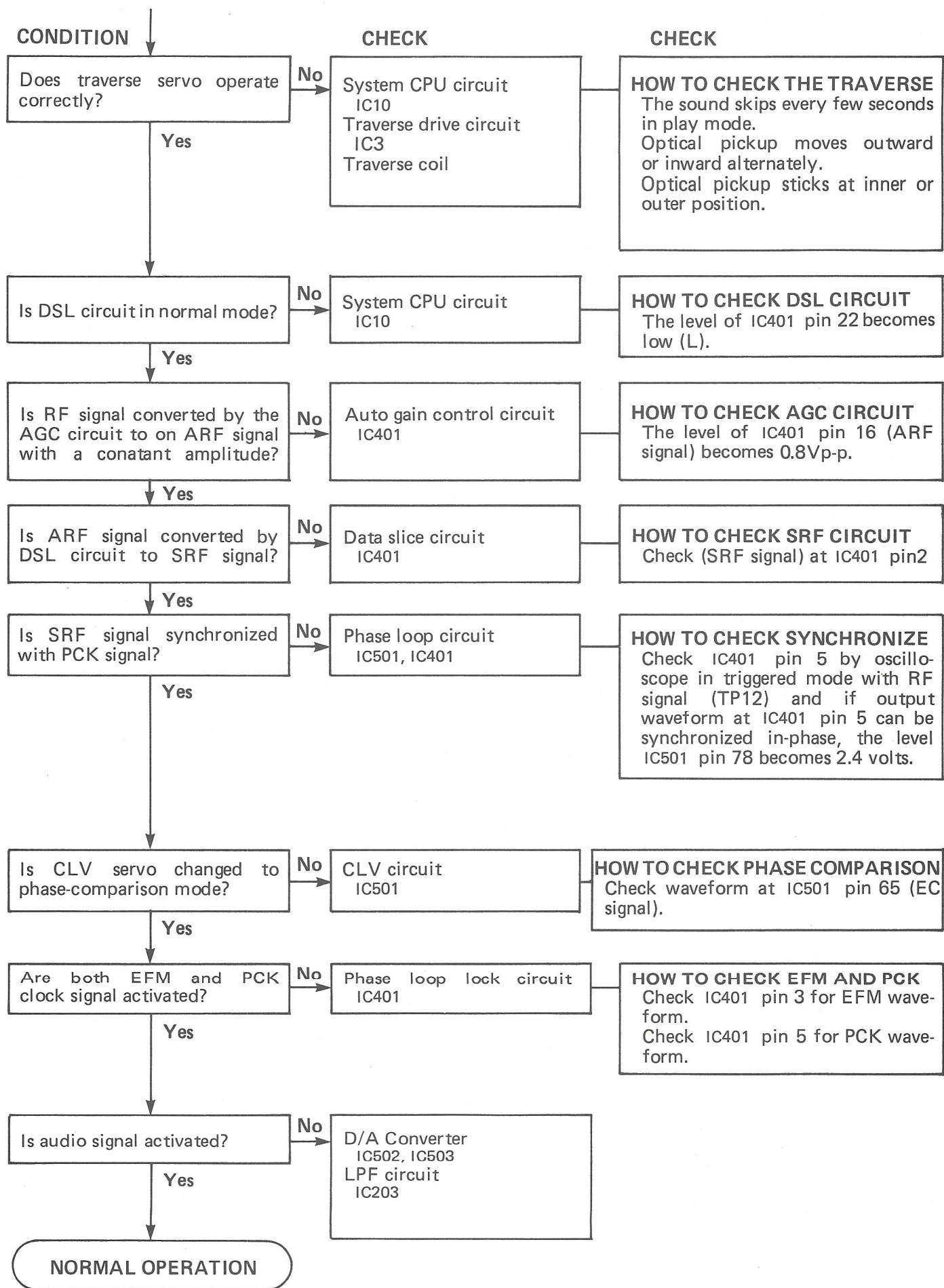


C6-3 TOC READ OPERATION – PLAY OPERATION FLOW CHART





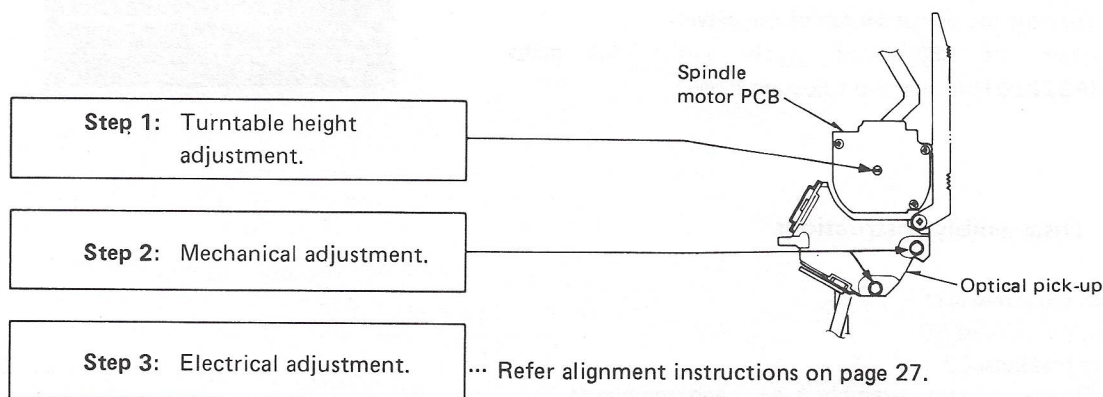
A5-2 OPTICAL PICKUP ADJUSTMENT INSTRUCTION (TYPE A)

A5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (SZZP1014F) new type CDT-002
Inspection test disc (SZZP1054C) CDT-D16
Uneven test disc (SZZP1056C) CDT-D18
- Hexagonal wrench (2mm)
- Screw lock paint (RZZ0L01)
- Lead wires
- Extension cord (Jig) YESFZ5020
(Supplied from APD) (See page 25.)
- Resistor (4.7 kΩ 1/4W)

A5-2-2 Adjust Procedure and Adjustment points

- If the optical pickup is replaced, adjust it according to the following procedure.

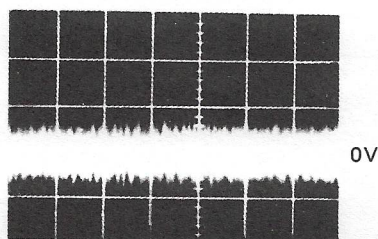
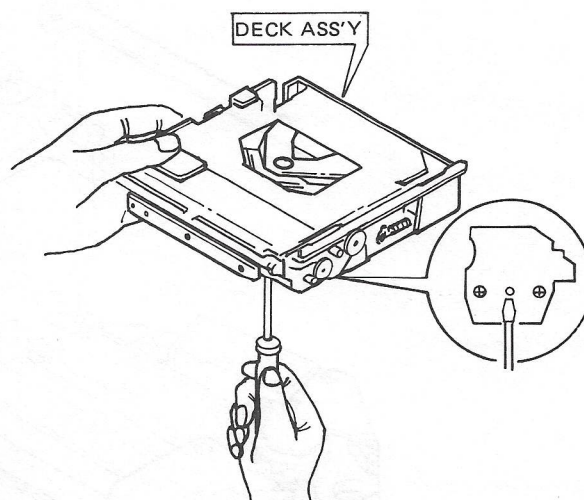
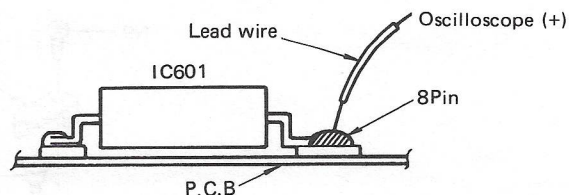


A5-2-3 Turntable Height Adjustment

- 1) Turn off the power of the player.
- 2) Solder a lead wire to IC601 Pin 8 of the SERVO PCB for oscilloscope connection.
- 3) Connect the extension cords between the Deck Ass'y and the unit.
- 4) Connect CH 1 of the oscilloscope to the IC601 Pin 8 (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
SWEEP 5 msec.
INPUT DC

- 5) Set the oscilloscope to DC zero balance.
- 6) Place the **test disc (SZZP1014F or SZZP1054C)** and turn on the power switch of the player.
- 7) Hold the CD Deck by the hand with the top face up.
- 8) The player to the play mode.
- 9) Turn the **adjustment screw** at the bottom of the spindle motor drive P.C.B. with a flat screwdriver so that the waveform is 0 ± 50 mV.
- 10) Turn off the power switch of the player.
- 11) After the adjustment, apply **screw lock paint (RZZ0L01)** to the adjusting screw.

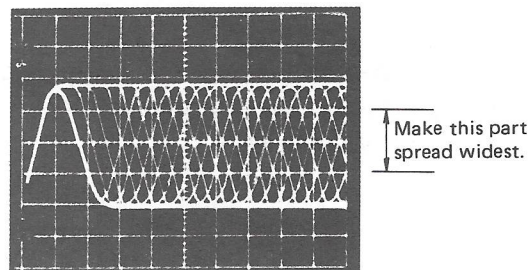
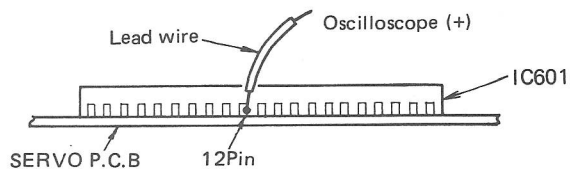


A5-2-4 Mechanical Adjustment

1. Connect CH1 of the oscilloscope to IC601 Pin 12 and Ground.

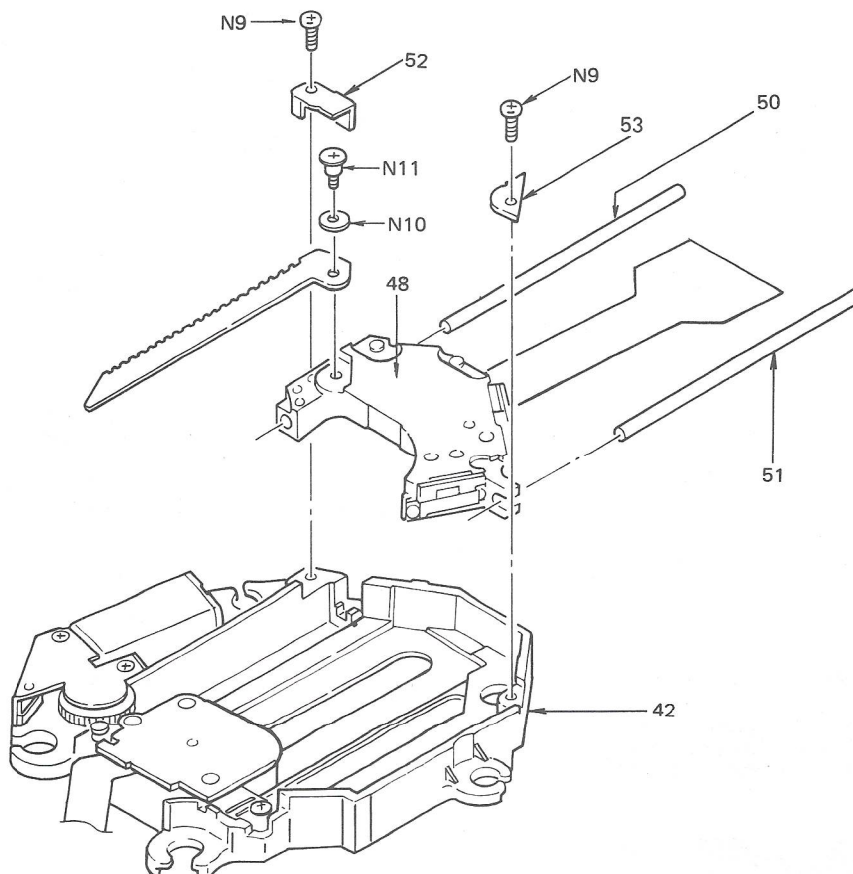
Oscilloscope setting: VOLT 200 mV
 SWEEP 0.5 μ sec.
 INPUT AC

2. Turn on the power of the player and play the Uneven Test Disc.
3. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
4. Turn off the power switch of the player.
5. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.

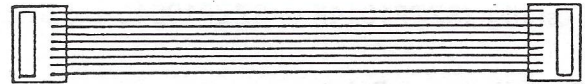


A5-2-5 Disassembly Instructions

1. Remove the screw N11.
2. Remove two screws N9.
3. Remove brackets 52 and 53
4. Raise Optical pickup assembly 48 and remove it.
5. Pull out the guide shafts 50 and 51 from the optical pickup assembly.



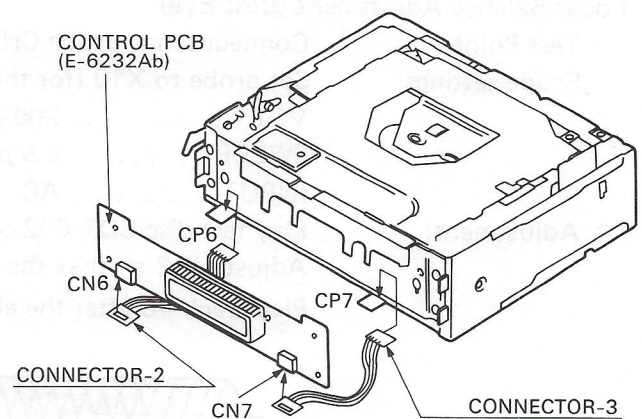
CONNECTOR-3 (8Pin)



CONNECTOR-4 (8Pin)



2) In the case of checking control PCB.

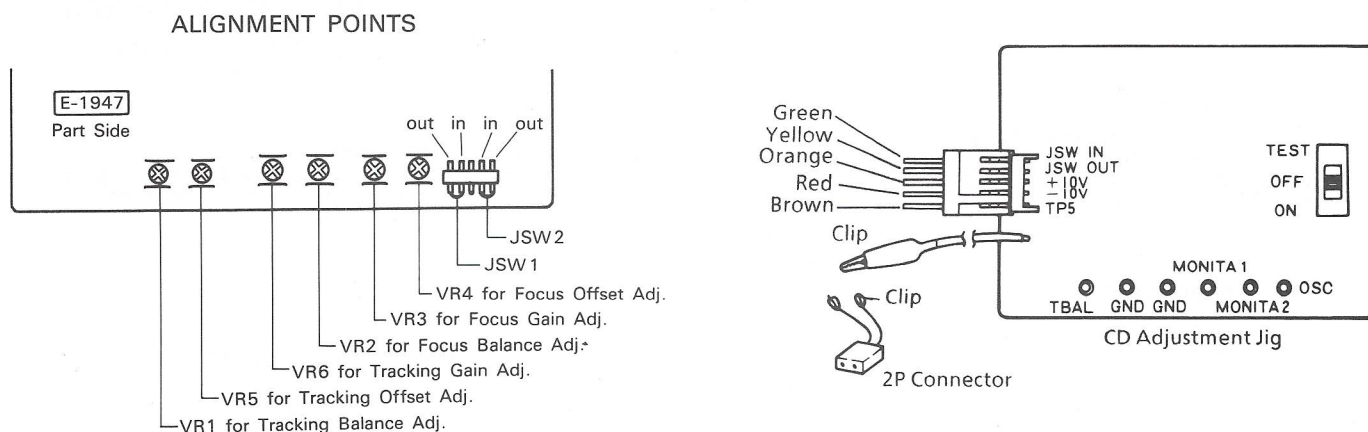


B5-1 ALIGNMENT INSTRUCTIONS (TYPE B)

B5-1-1 Measuring Instruments and Tools

- (1) CD adjusting jig W-2 YESFZ155
- (2) 2-channel oscilloscope
- (3) Low-frequency oscillator
- (4) Screwdriver for adjusting semi-fixed VRs
- (5) Test disc (SZZP1014F or SZZP1054C and SZZP1057C)

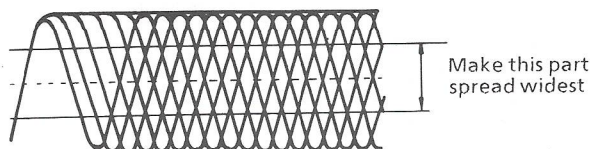
B5-1-2 Adjustment Points



B5-1-3 Electrical Alignment Procedures

1. Focus Balance Adjustment (Best Eye)

- Test Point: Connect scope probe CH1 to CN106 pin 8 or J58 on main PCB.
- Scope settings: Set probe to X10 (for the entire procedure)
VOLT 200—500 mV
SWEEP 0.5 μ s
INPUT AC
- Adjustment: Play test disc CDT-002 or CDT-016 track #1.
Adjust VR2 so that the eye pattern of RF signal is stretched to maximum.
Play track #6 after the alignment for confirmation check.



1. Focus Gain Adjustment (using test jig)

- Test Point: JSW2 (remove the short plug from JSW2)
- Connect the test jig (YESFZ155) as follow:

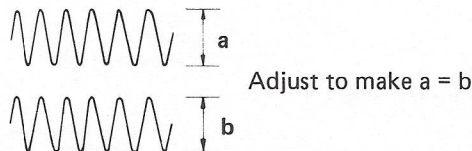
Clip	TO	CD chassis GND
Orange Lead (+10V)		CN401 pin 8 (+5V) on main PCB
Red Lead (-10V)		CN401 pin 3 (-5V) on main PCB
Green Lead (JSW in)		JSW2 out on Servo PCB
Yellow Lead (JSW out)		JSW2 in on Servo PCB

JSW2 (outside) Pin Identification JSW1 (inside)

On Servo PCB On Test Jlg	OUT IN (Green)	IN OUT (Yellow)	IN OUT (Yellow)	OUT IN (Green)
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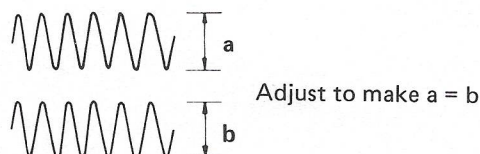
OSC Terminal TO Low Frequency Oscillator
Monitor 1 and 2 Scope CH1 and CH2

- Oscillator Settings: 1 kHz at 150 mV p-p
- Scope Settings: VOLT 100 mV
SWEEP 1 ms
INPUT AC
- Adjustment: Set the test jig level switch to TEST.
Play test disc (any track).
Adjust VR3 until both channels are equal in amplitude.



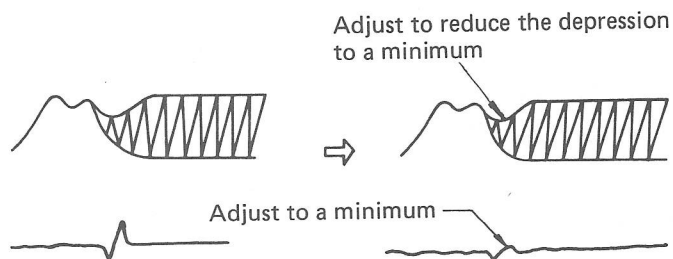
3. Tracking Gain Adjustment

- Test Point: JSW1 (remove the short plug from JSW1)
- All the connections and settings remain in the same as step 2 except JSW (green and yellow leads) are now connected to JSW1 on the Servo PCB (next to JSW2).
- Adjustment: Set the test jig level switch to TEST.
Play the test disc (any track).
Adjust VR6 until both channel are equal in amplitude.



4. Focus Offset Adjustment

- Test Point: Connect scope probe CH1 to CN106 pin 8 or J58 on main PCB.
Connect scope probe CH2 to JSW2 (with plug)
- Scope Settings: CH1 and CH2: VOLT 500 mV
SWEEP 0.5 ms (NOR)
INPUT AC
*** (CH1 set at normal trig) ***
- Adjustment: Play the Black Band test disc track #9.
Adjust VR4 to minimize the depression. Also make a fine adjustment of VR4 to make the FE signal of CH2 as flat as possible.

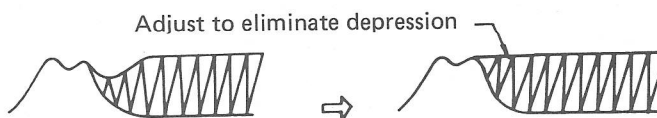


*** Continued in step 5 ***

5. Tracking Offset Adjustment

*** Continued from Step 4 ***

- Adjustment: Adjust VR5 to eliminate the depression. If necessary, adjust VR4 and VR5 alternately.



6. Tracking Offset Balance Adjustment (using test jig)

- Test Point: Connect the brown lead (TP5) from test jig to R3 (jump wire) or J44 behind VR1 on Servo PCB.
- Scope Settings: VOLT 100 mV
SWEEP 0.1 ms
INPUT DC
- Test Jig Connection: Clip TO CD chassis GND
Brown Lead (TP5) TP above
TBAL Terminal Scope Probe CH1
Green Lead (JSW in) JSW1 out (PCB)
Yellow Lead (JSW out) JSW1 in (PCB)
- Adjustment: Set the test jig level switch to ON.
Play the test disc track #1.
Adjust VR1 within 15 seconds after the test jig level switch is set to OFF so that the center of the waveform will be 0V.



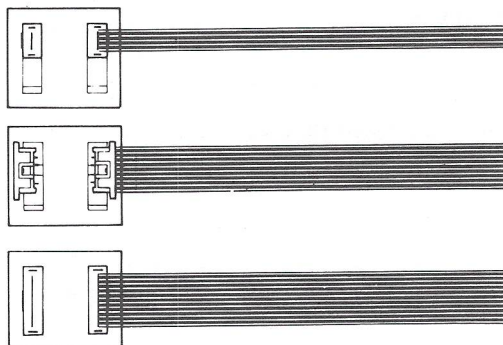
***** NOTE *****

1. If kicking noise is heard when playing the disc after having the alignments done, adjust the VR6 (tracking gain) for fine tuning until the kicking noise disappears.
2. Do not lose the JSW shorting plugs. Replace the shorting plug after each alignment step.
3. Turn off the power before changing any connection.

B5-2 OPTICAL PICKUP ADJUSTMENT

B5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (Supplied from HAD)
Test disc (SZZP1014F) new type CDT-002
Inspection test disc (SZZP1054C) CDT-016
Uneven test disc (SZZP1056C) CDT-018
- Screw lock paint (RZZ0L01) (Supplied from HAD)
- Hexagonal wrench (2 mm)
- Lead wires
- Extension cords
- Extension cords [YESFZ5003]
(Supplied from APD)



for CN4 (4P) 1PC

for CN1 (9P) 1PC

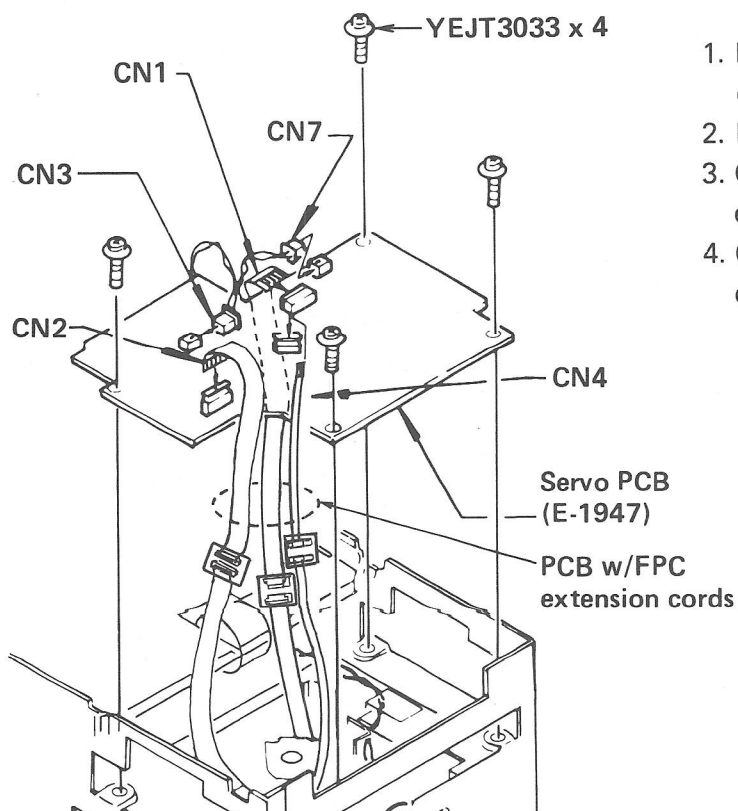
for CN2 (11P) 1PC

(2) Extension connectors



for CN3, CN7 (2P) 2PCS

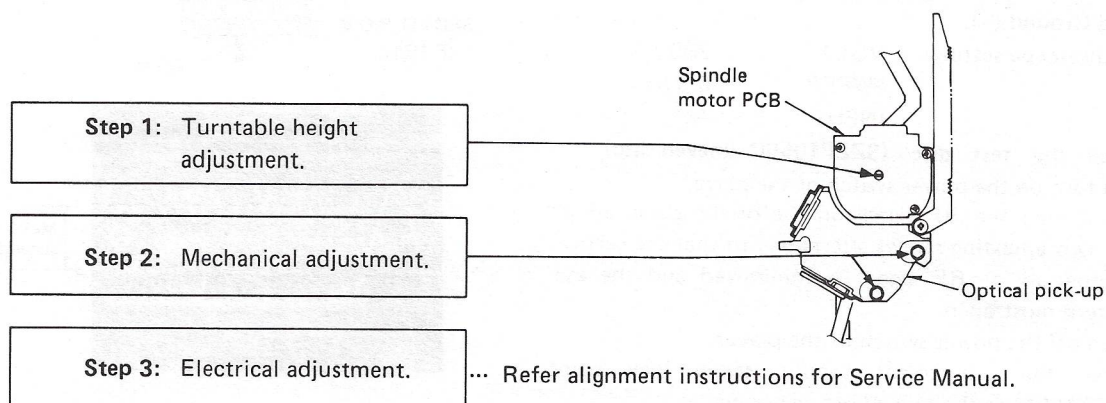
• Instruction for Jig



1. Remove the four ⊕ Screws and disconnect each connectors.
2. Remove the PCB (E-1947)
3. Connect each Jig connectors and FPC extension cords to optical pickup.
4. Connect each Jig connectors and FPC extension cords to Servo PCB (E-1947).

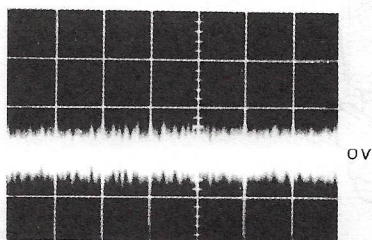
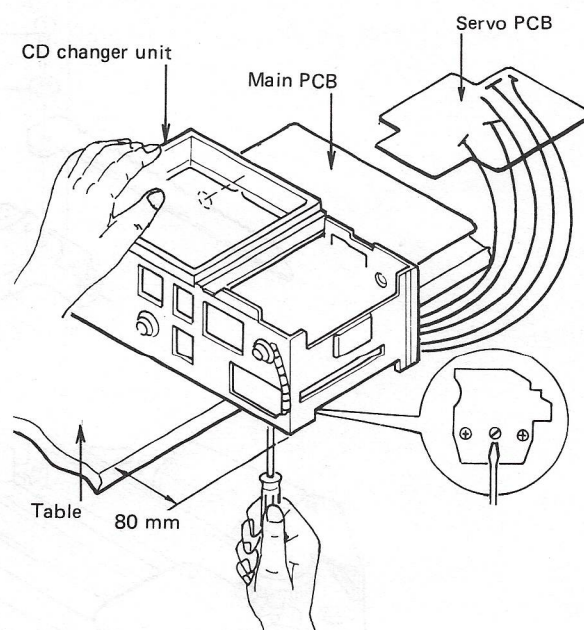
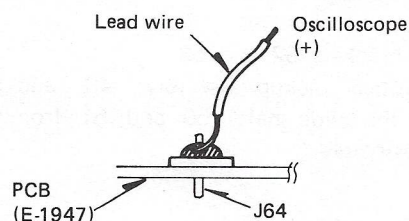
B5-2-2 Adjustment Procedure and Adjustment Points

- If the optical pickup is replaced, adjust it according to the following procedure.



B5-2-3 Turntable Height Adjustment

1. Turn off the power of the player.
2. Solder the lead wires to J64 (IC2, 37 pin) of the PCB (E-1947) and Ground so that the probe of oscilloscope can be connected.
3. Connect the extension cords between the Deck Ass'y (Refer instruction for Jig).
4. Put the CD changer unit in a posture that it is protruded about 80 mm from the edge of the table (repair servicing bench).
5. Connect CH1 of the oscilloscope to the J64 (+) and Ground (-) of the lead wires.
Oscilloscope setting: VOLT 200 mV
SWEEP 5 m sec.
INPUT DC
6. Set the oscilloscope to DC zero balance.
7. Place the test disc (SZZP1014F or SZZP1054C) and turn on the power switch of the player.
8. The player to the play mode.
9. Turn the adjusting screw at the bottom of the spindle motor drive PCB with a flat screwdriver so that the waveform is 0 ± 50 mV.
10. Turn off the power switch of the player.
11. After the adjustment, apply screw lock paint (RZZ0L01) to the adjusting screw.

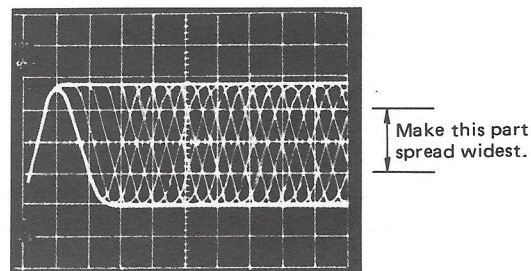
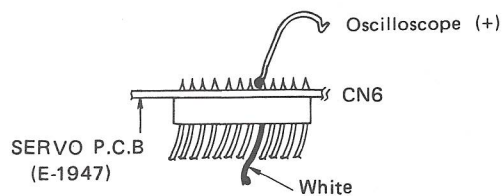


B5-2-4 Mechanical Adjustment

1. Solder a lead wire to CN6 for 8 pin (white) of the SERVO PCB (E-1947).
2. Connect CH1 of the oscilloscope to the lead wire (+) and Ground (-).

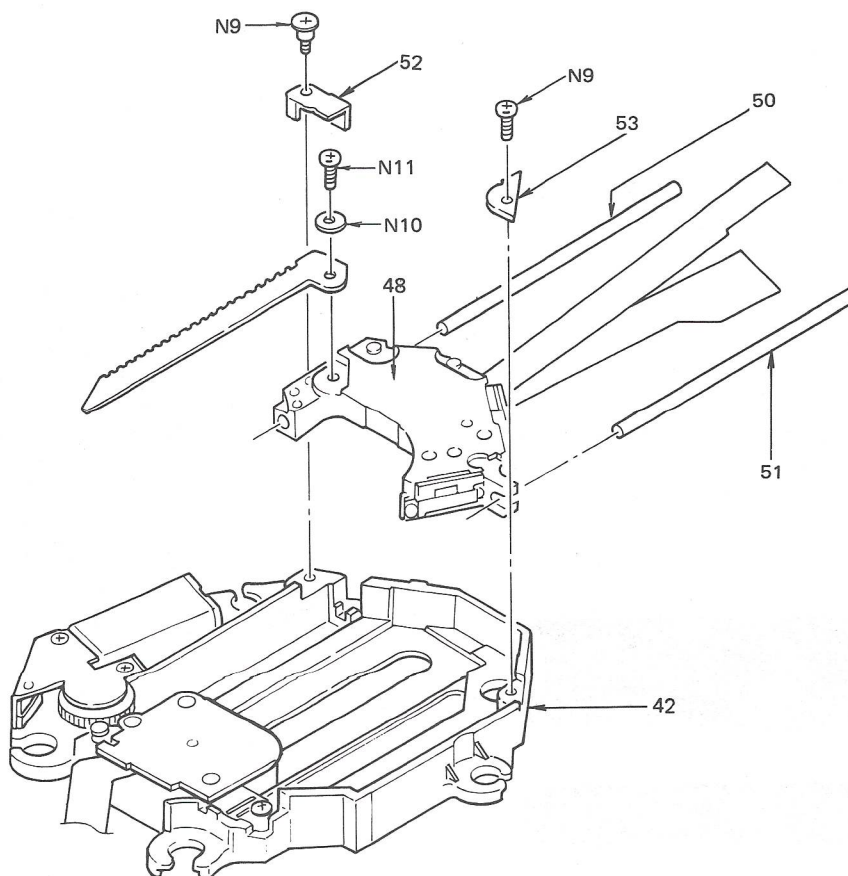
Oscilloscope setting: VOLT 200 mV
SWEEP 0.5 μ sec.
INPUT AC

3. Place the test disc (SZZP1056C uneven disc) and turn on the power switch of the player.
4. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
5. Turn off the power switch of the player.
6. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.



DISASSEMBLY INSTRUCTIONS

1. Remove the screw N11.
2. Remove two screws N9.
3. Remove brackets 52 and 53.
4. Raise Optical pickup assembly 48 and remove it.
5. Pull out the guide shafts 50 and 51 from the optical pickup assembly.



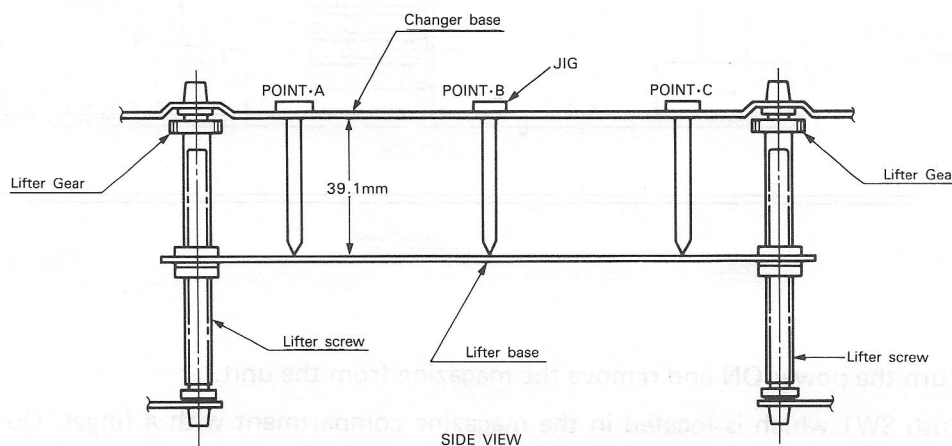
B5-3 MECHANICAL ALIGNMENT PROCEDURES:

Test Fixtures Required:

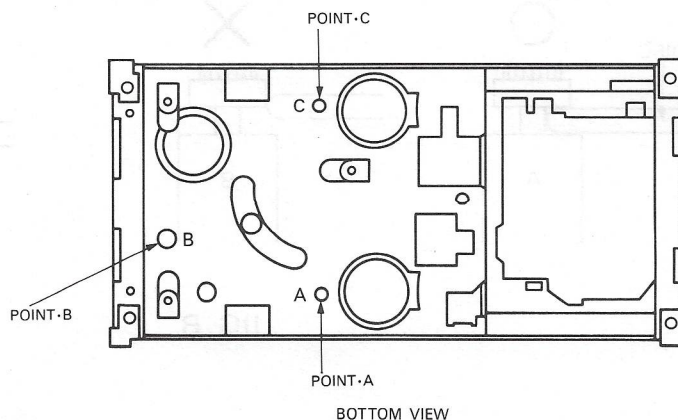
- YESFZ5001 — Lifter Base Height Adjustment Jig (B)
- YESFZ5002 — Lifter Base Level Adjustment Jig (A)

1. Lifter Base^A Level Adjustment (Perform this adjustment first)

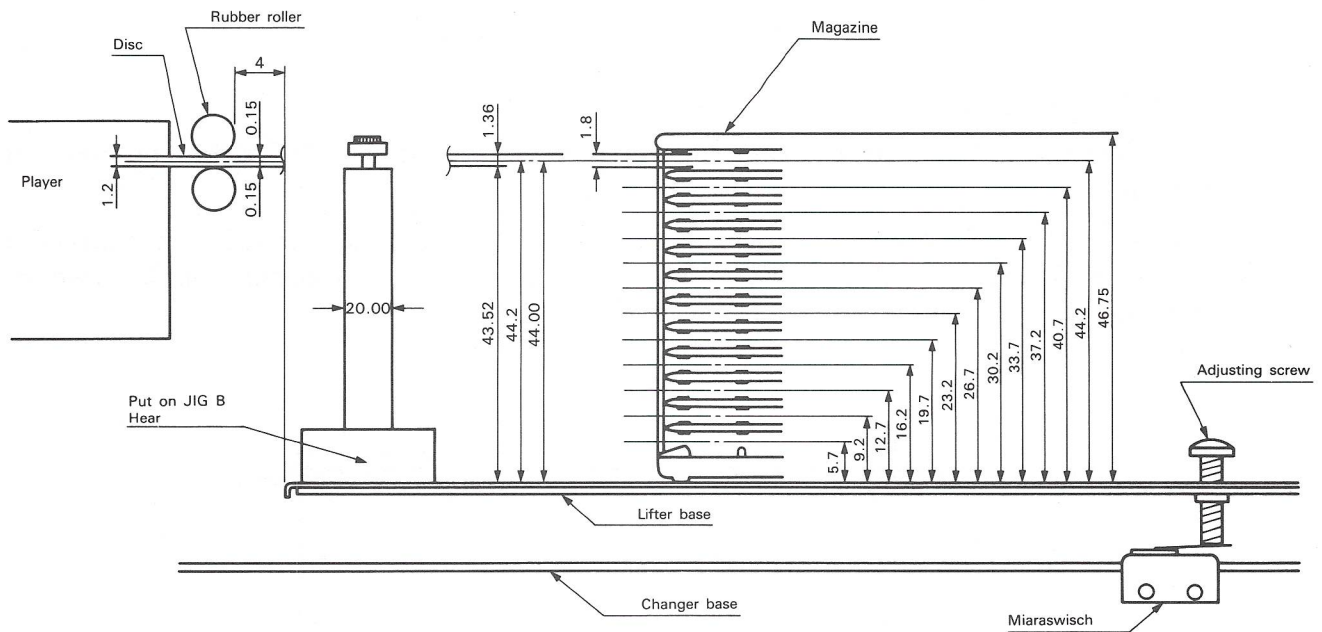
- Turn the power ON and tune to Disc 9 position by pressing the "Disc > " buttons on the remote control.
- Turn the power OFF and insert the Jig A into points A, B and C alternately. Both the tip and the head of Jig A should touch the bottom side of the lifter base and the bottom chassis respectively at all points.



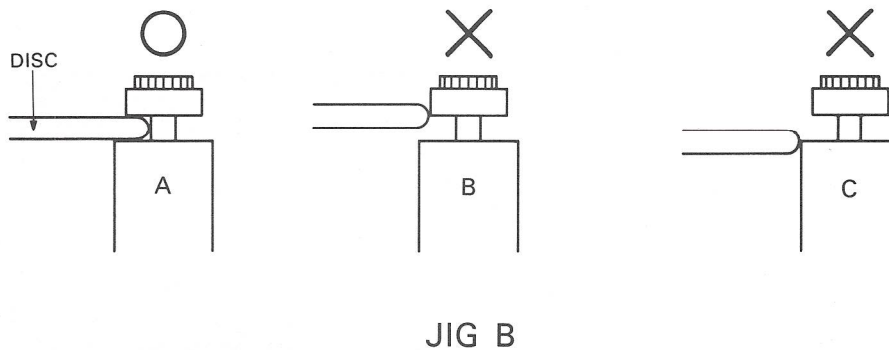
- If either tip or the head of the Jig A does not touch the respective locations, the adjustment will begin with removing all 3 E-rings, ref. No. 245, and 3 extension gears, ref. No. 281.
- Turn the lifter gears, ref. No. 282, either clockwise or counterclockwise until both tip and head are touched by respective locations at all points.
- Replace the E-rings and the extension gears after completion of the adjustment.



2. Lifter Base Height Adjustment



1. Turn the power ON and remove the magazine from the unit.
2. Push SW1 which is located in the magazine compartment with a finger. Do not release until the lifter base completely reaches to the bottom and stop.
3. Press the "Play" button and load a disc into the disc compartment. Disc will begin to play.
4. Press the "Eject" button and the disc will eject.
5. Turn the power OFF when the disc is half way out.
6. Place the Jig B toward the disc and you will note that one of the diagrams listed below will be matched.



7. If diagram A is matched, no adjustment is needed.
8. If diagram B is matched, turn the adjusting screw quarter-turn clockwise.
9. If diagram C is matched, turn the adjusting screw quarter-turn counter-clockwise.
10. After the screw adjustment, turn the power ON and remove the disc from the unit. Let the lifter base to elevate to the top and stop.
11. Repeat steps 2–6 above to check if the proper alignment is made. Otherwise, repeat step 8 or 9 until adjustment matches diagram A.
12. After obtaining the correct height adjustment, repeat steps 2–6 one more time for confirmation.

C5-1 ALIGNMENT INSTRUCTION (TYPE C)

C5-1-1 Measuring Instruments and Tools

1. CD adjusting jig YESFZ155
2. Two-channel oscilloscope
3. Low-frequency oscillator
4. Screwdriver for adjusting semi-fixed variable resistors
5. Extension Cord Kit 3 pcs YESFZ155
6. Test Disc (SZZP1014F or SZZP1054C and SZZP1057C)

"Caution-use of controls or adjustments or performance of procedures other than those specified therein may result in hazardous radiation exposure."

C5-1-2 Alignment Procedures and Points

Remove the top case and bottom case, solder the leads to TP and JSW on the back of the circuit board, and follow the sequence below.

Note: Do not attempt alignments when the deck is removed.

- 1) Focus offset alignment VR105 (TP3)
- 2) Tracking offset alignment VR106 (TP1)
- 3) Focus balance alignment VR103 (CN4 for Pin 4 (YEL))
- 4) Tracking balance alignment VR107 (JSW1 both ends of terminals)
- 5) Focus gain alignment VR104 (JSW2)
- 6) Tracking gain alignment VR102 (JSW1)
- 7) Focus retra alignment VR105 (TP3, CN4 for pin 4(YEL))
- 8) Tracking alignment VR106 (TP1, CN4 for Pin 4(YEL))

After adjustment the above, disconnect the leads, return the JSW short ties as they were before, and check performance.

Additional Adjustment (PLL)

Remove the deck and connect the PLL with a connecting cable for PLL adjustment.

- ※) Additional PLL adjustment VR401 (Pin ②, IC501)

C5-1-3 Adjusting Methods

1) Focus offset alignment

- Test point: TP3
- Oscilloscope setting: Volt 100mV Sweep 1ms Input selector DC
- Operation: Open the door without inserting a disc.
- Alignment: Adjust VR105 until the DC voltage is 0V.
- Waveform sample:



2) Tracking offset alignment

- Test point: TP1
- Oscilloscope setting: Volt 100mV Sweep 1ms Input selector DC
- Operation:
 - a) Insert a disc and play the outermost track.
 - b) During play, hold the disc with fingers to stop the rotation.
- Alignment: Adjust VR106 until the center of the waveform displayed on the scope is 0V.
- Waveform sample:



3) Focus balance alignment (Best eye)

- Test point: Connect the Scope to CN4 for pin 4 (Yellow) and GND.
- Oscilloscope setting: Volt 50–100mV 20mV Sweep 0.5μs Input selector AC
- Operation: Insert disc and play track #1 for alignment, track #6 after the alignment.
- Alignment: RF signal appears. Adjust VR103 until the eye pattern extends widest.
- Waveform sample:



4) Tracking balance alignment

- Test point: JSW1 both ends of terminals, TP1
- CD adjusting jig connection

1) Clip	Chassis
2) Lead. TP1 (brown)	TP1
3) TR BAL terminal	Oscilloscope probe
4) Lead JSW IN (green), lead JSW OUT (yellow)	Both ends of JSW1 (Remove short tips)
(Top pin) out, (bottom pin) in when deck at upside down position	(Remove short tips)
- Oscilloscope setting: Volt 100mV Sweep 0.1ms Input selector DC
- Operation: Insert a disc and play track #1. Set the lever switch on the CD adjusting jig to on.
- Alignment: Shift the lever switch to OFF, and adjust VR107 so that the center of the waveform will be 0V with in 15 seconds.
- Waveform sample:



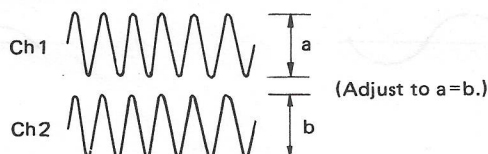
5) Focus gain alignment

- Test point: Both ends of JSW2 (Remove short plug) (Top pin) IN, (bottom pin) OUT when deck at upside down position.
- CD adjusting jig connection

1) Clip	Chassis
2) Lead +10V (orange)	+10V (E of Q109)
3) Lead -10V (red)	-10V (E of Q108)
4) Lead JSW IN (green)	JSW2 IN
5) Lead JSW OUT (yellow)	JSW2 OUT
6) OSC terminal:	Low frequency OSC
7) Oscilloscope Ch1, Ch2 terminals	Oscilloscope probe
- Low frequency oscillator setting

1) Frequency	600Hz ~ 1 kHz
2) Level	150mV p.p.
- Oscilloscope setting (Ch1, Ch2):

Volt 200mV	Sweep 1ms	Input selector AC
--------------------	-------------------	---------------------------
- Operation: Set the lever switch on the CD adjusting to TEST Insert a disc and play.
- Adjust VR104 until the level of both channels at 1kHz are the same.



6) Tracking gain alignment (Use of Adjusting jig)

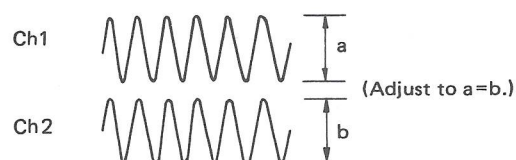
- Test point: Both ends of JSW1 (Remove short tips) (Top pin) OUT (bottom pin) IN
- CD adjusting jig connection:

1) Clip	Chassis
2) Lead +10V (orange)	+10V (E of Q109)
3) Lead -10V (red)	-10V (E of Q108)
4) Lead JSW IN (green)	JSW1 IN
5) Lead JSW OUT (yellow)	JSW1 OUT
6) OSC terminal	Low frequency OSC
7) Monifor1, Monifor2 teminals	Oscilloscope probe Ch1, Ch2
- Low frequency oscillator setting

1) Frequency	1 kHz ~ 1.5 kHz
2) Level	1 50mV p-p
- Oscilloscope setting (Ch1, 2)

Volt 200mV	Sweep 1ms	Input selector AC
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- Operation: Set the lever switch on the CD adjusting jig to TEST. Insert a disc and play.
- Alignment: Adjust VR102 until the levels of both channels are the same.

- Waveform sample:



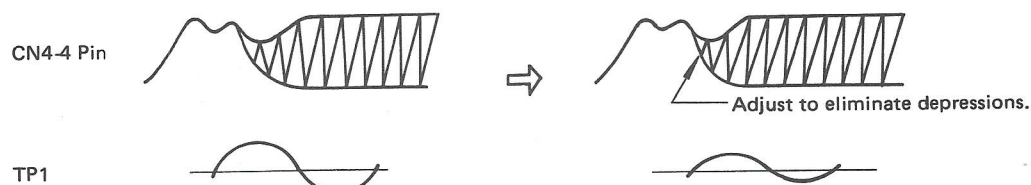
7) Focus retra alignment

- Test point: Connect Pin 4 of CN4 to Ch1; TP3 to Ch2.
- Oscilloscope setting (Ch1, 2): Volt 50mV (CH1), 500mV (CH2) Sweep 0.5ms (Nor.) (CH1)
Input selector AC (CH1) DC (CH2)
- Operation: Insert a test disc and play track #B. Reproduce a black band of 0.5mm.
- Alignment: a) Watch the eye pattern for Ch1, and make fine adjustment of VR105 to a minimum depression.
b) Also make a fine adjustment of VR105 so that FE signal for CH2 will be as flat as possible.
- Waveform sample:



8) Tracking offset alignment

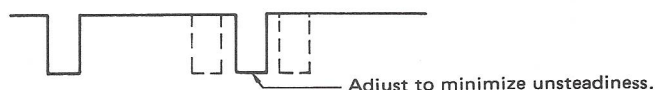
- Test point: Connect the Scope to CN4 for pin 4 (CH1) TPI (CH2)
- Oscilloscope setting (Ch1, 2) Volt 500mV Sweep 0.5ms Input selector AC
- Operation: Insert a test disc and play. Reproduce a black band of 0.5mm.
- Alignment: a) Watch the eye pattern for Ch1, and make fine adjustment of VR106 to a eliminate depression



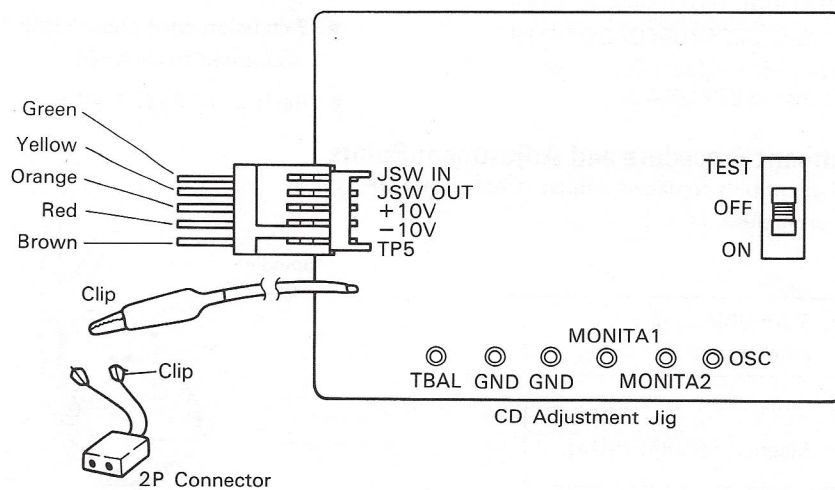
※) Additional alignment PLL adjustment

note: This alignment cannot be made unless the deck is taken out.

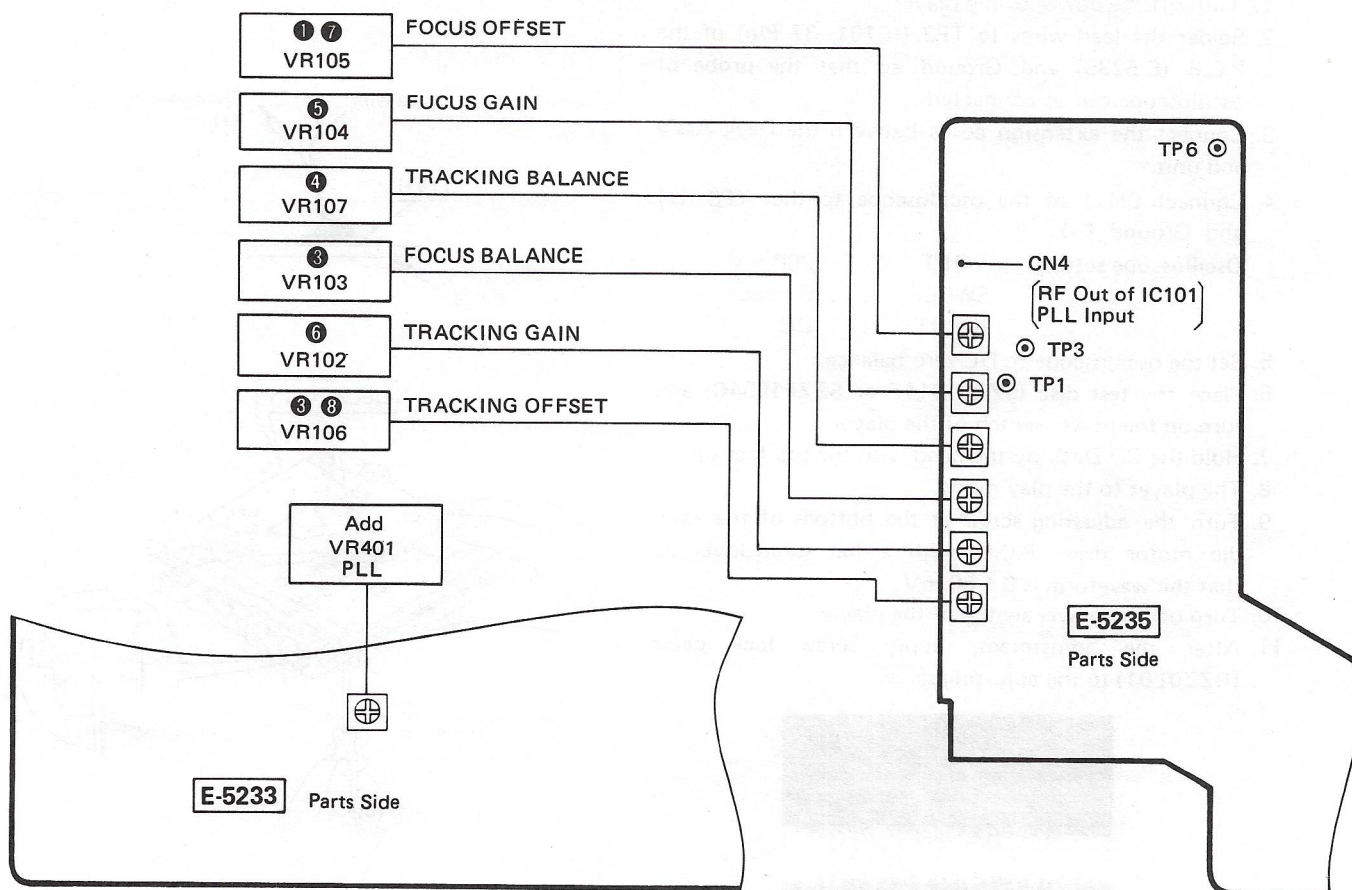
- Test point Pin ② (CLDCK) of LSI (MN6617)
- Oscilloscope setting: Volt 2.0V Sweep 50μs Input selector DC Trig (NOR)
- Operation: Insert a test disc and play. Play track #8. Reproduce a black band of 0.5 mm.
- Alignment: Adjust VR401 until the waveform is steady or as steady as possible.
- Waveform sample: Adjust to minimize unsteadiness.



C5-1-4 CD Adjustment Jig



C5-1-5 Alignment Points



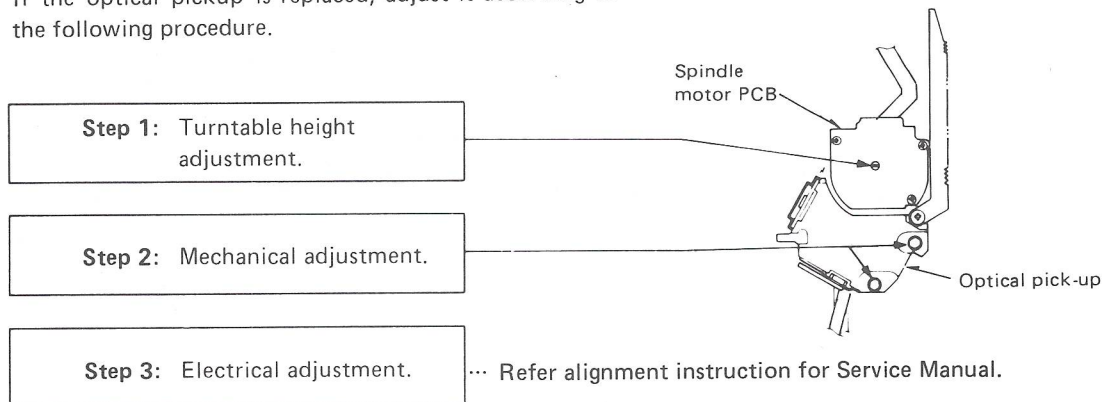
C5-2 OPTICAL PICKUP ADJUSTMENT

C5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (SZZP1014F) new type CDT-D02
Inspection test disc (SZZP1054C) CDT-D16
Uneven test disc (SZZP1056C) CDT-D18
Inspection test disc (SZZP1054C)
- Hexagonal wrench (2 mm)
- Screw lock paint (RZZ0L01)
- Lead wires
- Extension cord (Jig) [YESFZ154] (Supplied from APD)
- Resistor (4.7 k Ω 1/4W)

C5-2-2 Adjustment Procedure and Adjustment Points

- If the optical pickup is replaced, adjust it according to the following procedure.

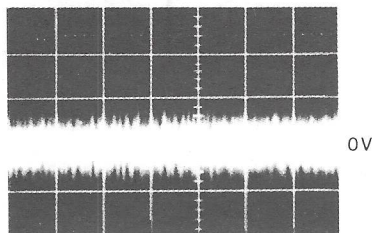
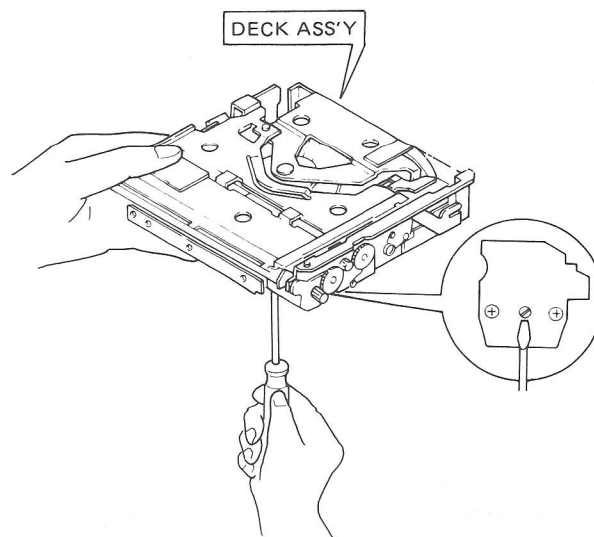
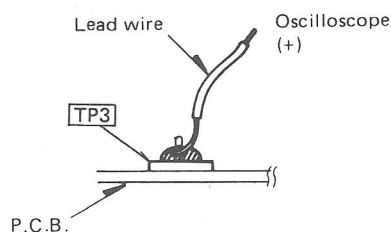


C5-2-3 Turntable Height Adjustment

1. Turn off the power of the player.
2. Solder the lead wires to TP3 (IC101, 37 Pin) of the P.C.B (E-5235) and Ground so that the probe of oscilloscope can be connected.
3. Connect the extension cords between the Deck Ass'y and unit.
4. Connect CH 1 of the oscilloscope to the TP3 (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
SWEEP 5 msec.
INPUT DC

5. Set the oscilloscope to DC zero balance.
6. Place the **test disc (SZZP1014F or SZZP1054C)** and turn on the power switch of the player.
7. Hold the CD Deck by the hand with the top face up.
8. The player to the play mode.
9. Turn the **adjusting screw** at the bottom of the spindle motor drive P.C.B. with a flat screwdriver so that the waveform is 0 ± 50 mV.
10. Turn off the power switch of the player.
11. After the adjustment, apply **screw lock paint (RZZ0L01)** to the adjusting screw.

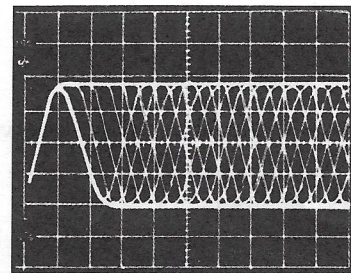
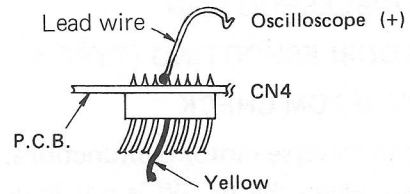


C5-2-4 Mechanical Adjustment

1. Solder a lead wire to Pin 4 of CN4 (Yellow) of the P.C.B.
2. Connect CH1 of the oscilloscope to the lead wire (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
 SWEEP 0.5 μ sec.
 INPUT AC

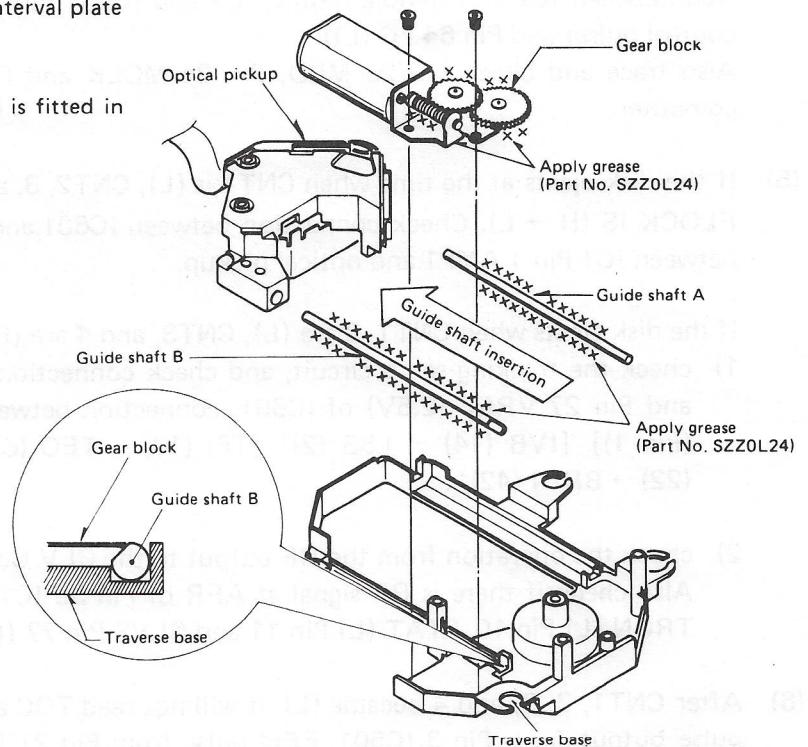
3. Place the uneven test disc and turn on the power switch of the player.
4. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
5. Turn off the power switch of the player.
6. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.



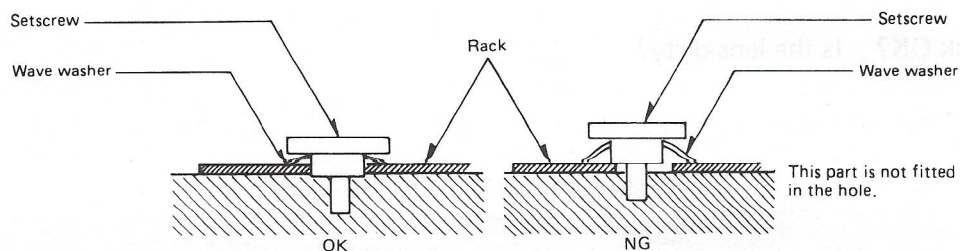
Cautions for assembly

The assembly work can be done by reversing the disassembly procedure. Keep the following points in mind.

1. Be sure to insert the guide shaft in the direction of the arrow.
 If it is inserted in the wrong direction, the interval plate spring may be damaged.
2. Make sure that the D cut off guide shaft B is fitted in the position as shown.



3. Tighten the rack setscrew as shown.



6. TROUBLESHOOTING

A6 TROUBLESHOOTING (TYPE A)

A6-1 SYMPTOM CHECK

- (1) When traverse motor malfunctions. Check IN SW (H).
Also, when the IN SW is not in changing stage, check the system CPU Pins 18 and 19 (TRV R, TRV F), and Pins 14–17 (CNT1, 2, 3, 4, all H) outputs to IC601, then from IC601 (TRV F/R) to IC602, from IC602 to the traverse motor.
- (2) When the focus lens moves up and down repeatedly, check if the disk is properly set. If the LD ON terminal of Pin 28 IC601 is (H), check the LD terminal of Pin 7 IC1. To see if APC circuit is turned on to intensify the laser.
Check connections between the photo-detector and IC1, check if there is a pulse at the FE terminal of Pin 15 IC1, and a pulse at SENSE terminal of Pin 24 IC601.
- (3) If the disk ejects, when CNT1 is (H), CNT2, 3, and 4 are (L). Check to see if the CLW terminal of Pin 10 IC601, the FPO terminal of Pin 24 IC1, and FD terminal of Pin 10 IC602 generate the triangular wave respectively.
- (4) If the disk ejects at the time when the disk detection ended (CNT1, 2, 3, and 4 (H)), check connection between IC2 and spindle motor, IC2 and IC501 Pin 32 L/R (Clock pulse), Pin 65 EC (Start-up control pulse) and Pin 64 PC (L)).
Also trace and check Pin 23 MLD, Pin 24 MCLK and Pin 25 MDATA (data pulse) of the micro-computer.
- (5) If the disk ejects at the time when CNT1 is (L), CNT2, 3, and 4 are (H), check to see if IC601 Pin 26 FLOCK IS (H → L). Check connection between IC601 and IC1 Pin 32 RFDET (L), Pin 31 BDD and between IC1 Pin 1 AMP1 and optical pickup.

If the disk ejects when CNT1, 2 are (L), CNT3, and 4 are (H),

- 1) check the tracking servo circuit, and check connection between tracking coil and Pin 41 IC601, and Pin 27 VREF (2.5V) of IC601, connection between IC1 and IC601 [IVA (Pin 13) → LSA (Pin 1)] [IVB (14) → LSB (2)] [TPI (17) ← TEO (6)] [TGC (25) (0V) ← VDET (23)] [TPO (22) → BRIN (42)].
 - 2) check the operation from the RF output to the CLV Control.
Also check if there is RF signal at AFR of Pin 35 IC1, at SRF of Pin 19 IC601 and also check TRON (L) Pin 10, STAT (L) Pin 11 and CLVS Pin 77 (L → H) of IC501.
- (6) After CNT1, 2, 3, and 4 became (L), it will not read TOC and the disk ejects, check if there is SUBQ pulse output from Pin 3 IC501, EFM pulse from Pin 21 IC601. Check KICK at Pin 27 IC601 and TVPO output at Pin 39 IC601. If SUBQ is output the AMS (auto-music scan) is operated by the Kick Pulses then read TOC in the micro-computer.

* Is the disk OK? Is the lens dirty?